Power

Power is rate of doing work. Power is work done per second or energy transferred per second Power is given by the following relationship:

$$Power\ P = \frac{work\ done}{time\ taken} = \frac{energy\ transferred}{time} = \frac{W}{t}$$

- P is measure in watts
- W is measures in joules
- t is measured in seconds
- A power of 1 watt means that 1 joule of energy is being transferred every second.

Example question

1. A person pedalling on an exercise bike does 1200 J of work in 5 s.

a) How much energy is transferred by the person in 5 s?
b) What is the power of the person during the 5 s of the exercise?

Answers:

a) energy transferred in 5 s

$= work\ done$
$= 1200\ J$

b) power of the person

$$= \frac{energy\ transferred}{time}$$
$$= \frac{1200}{5}$$
$$= 240W$$

Practice questions:

1. A set of pulleys lifts a piano and does 3,356 joules of work. If the piano is lifted in 75 seconds, how much power is used?
2. How much work is done in order to cook a bag of popcorn in a 500 watt microwave oven for 5.5 minutes?
3. What is the power of an electric toothbrush if it can do 755.8 joules of work in 75 seconds?
4. Sara and Josh do the exact same amount of work. Sara does the work in 2.3 hours and Josh does it in 2.5 hours. Who is more powerful? Explain.
5. A dock worker lifts a 375N crate a distance of 0.5m over his head in 2.3 seconds. What is his power output?

- Hint: You must do a calculation before you can begin to solve for *power*.

6. An athlete is using the row machine in the gym. She does 3245 joules of work on the oars in 72 seconds. What is her power output?
7. John is walking to class up the front staircase. He weighs 595N and the stairs are 3.15 meters long. Calculate his power output if he climbs the stairs in 15.2 seconds.